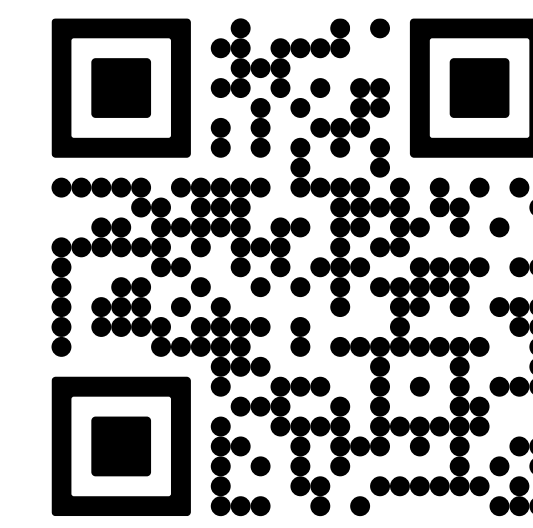




Asymptotic Analysis as an Abstract Interpretation

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GitHub Repo

Motivation & Contribution

Motivation

- Abstract interpretation (AI): Powerful theory of semantic approximation, but **mathematically heavy**
 - Asymptotic analysis (AA): **Familiar to undergraduates**
- ⇒ Formalize AA as an **intuitive example** of AI!

Our contribution

- ✓ **Minimal formalization** without complete partial orders (CPOs) or fixed points
 - ✓ Preserves **core theoretical structure** of AI
 - **Galois connection**: guarantees formal soundness
 - **Assertion checking**: safe approximations with proof
 - ✓ **Partial mechanization** of proofs in Lean 4
 - ✓ **Interactive tools** for exploration (interpreter, fuzzer)
- ⇒ Pedagogical pathway to program analysis

Impacts

- Makes AI **accessible** to undergraduates
- Provides **executable tools** for hands-on learning
- Opens path to **lightweight analyses** for pedagogy

Leading-Term Analysis (LTA)

Leading-term analysis

- Retaining only the term with the **largest exponent**
 - (i.e., the dominating term for $n \rightarrow \infty$)
- ⇒ Discards lower-order detail
& Captures asymptotic behavior

e.g.) Complicated polynomial

$$f(n) = \underbrace{(n+1)}_n \underbrace{(3n+1)}_{3n} - n \approx 3n^2$$

Diagram illustrating the simplification of a polynomial expression by focusing on the dominant term $3n^2$.

Why LTA, not big-O?

- ✓ Preserves **coefficients**: Distinguishes $2n^2$ from $100n^2$ (equivalent treatment with big-O)
⇒ **Richer, quantitative analysis**
- ✓ Aligns more well with **pre-calculus intuition**
- ✓ **Observable precision loss**
 - Term omission: $n^2 + n \rightarrow n^2$
 - Cancellation: $n^2 - (n + n^2) \rightarrow ?$

Language of Natural-to-Real Functions

Grammar

$P ::= n^r$	Monomial ($r \in \mathbb{R}$)
$ 2^n$	Exponential
$ cP$	Scale ($c \in \mathbb{R}$)
$ P \pm P$	Sum / Difference
$ P \cdot P$	Product
$ \max\{ P , P \}$	Maximum (of absolute values) (Compatible with abstract ordering)

- Motivated by arithmetic expressions
(Often defined in introductory PL courses)
- Subtraction: Allows for **overapproximation** later
- Max: Frequently used for **algorithm analysis**
(e.g., parallel computing)

Formalization

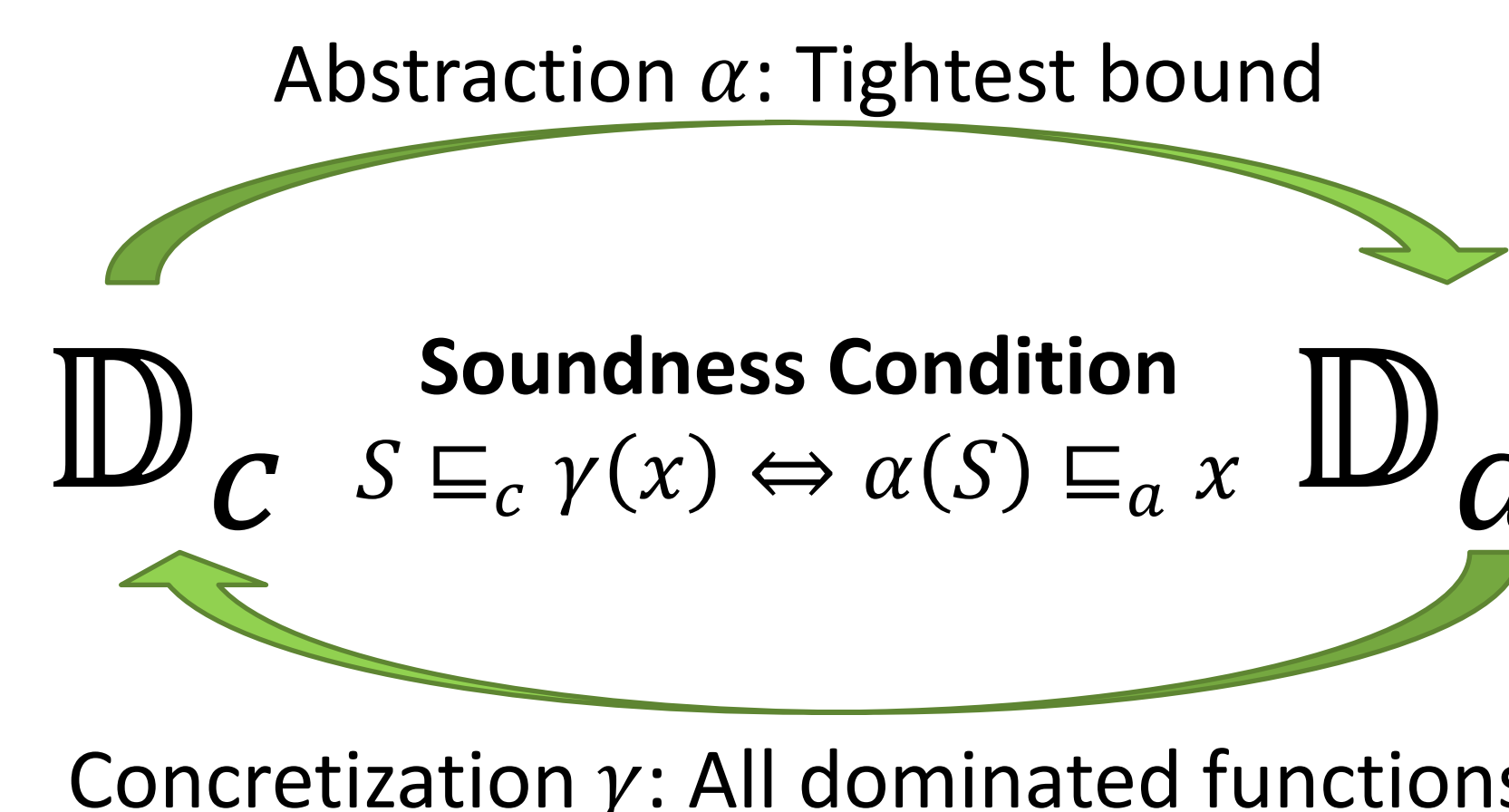
Semantic domains and orderings

- Denotational domain: $\mathbb{D}_d = \mathbb{R}^{\mathbb{N}}$
 - $f \sqsubseteq_d g$ iff $|g(n)|$ grows faster than $|f(n)|$
 - Concrete domain: $\mathbb{D}_c = \mathcal{P}(\mathbb{D}_d)$
 - Subset ordering $\sqsubseteq_c = \subseteq$
 - Abstract domain: $\mathbb{D}_a = [0, \infty] \times \mathbb{R}$
 - (c, r) denotes the growth rate of cn^r
 - Lexicographic ordering:
 $(c_1, r_1) \sqsubseteq (c_2, r_2)$ iff
 $r_1 < r_2$ or $(r_1 = r_2 \wedge c_1 \leq c_2)$
- ✓ Lean 4 mechanization of ordering properties

Abstract domain elements

- $\perp$: Subpolynomial
e.g.) $0, 2^{-n}$
 - $(0, r)$: Slower than n^r , but faster than n^s for $s < r$
e.g.) $n^r / \log n$
 - (c, r) : Growth rate of cn^r where $c \in (0, \infty)$
e.g.) $3n^2, 1/n$
 - (∞, r) : Faster than n^r , but slower than n^s for $s > r$
e.g.) $n^r \log n$
 - T: Superpolynomial
e.g.) 2^n
- (Note: $2^{-n}, \log n$ are possible extensions to the language.)

Galois connection



- ✓ Proof given in extended abstract
- ❖ Lean 4 mechanization in progress

Overapproximation from total term cancellation

e.g.) $P = \underbrace{n^2}_{(1,2)} - \underbrace{(n+n^2)}_{(1,1)(1,2)}$

$\llbracket P \rrbracket_a = (0, 2)$

True leading term: $-n \rightarrow (-1, 1) \sqsubset_a (0, 2)$

⇒ Cancellation causes **imprecise bound**!

Demonstrates:

- ✓ Tangible loss of information (term cancellation)
- ✓ Trade-off: precision vs. tractability
- ✓ Pedagogical opportunity for improvement

Assertion Checking

Soundness theorem

The following assertions are sound:

- If: $\llbracket P \rrbracket_a = \perp$
Then: $\llbracket P \rrbracket_d(n)$ grows slower than any polynomial
Formally: $(\forall r \in \mathbb{R}) \lim_{n \rightarrow \infty} \frac{\llbracket P \rrbracket_d(n)}{n^r} = 0$
- If: $\llbracket P \rrbracket_a = (c, r)$ (leading term cn^r)
Then: $|\llbracket P \rrbracket_d(n)|$ grows at most as fast as $|c|n^r$
Formally: $\limsup_{n \rightarrow \infty} \left| \frac{\llbracket P \rrbracket_d(n)}{n^r} \right| \leq |c|$
- If: $\llbracket P \rrbracket_a = (\infty, r)$
Then: $\llbracket P \rrbracket_d(n)$ grows slower than every n^s for $s > r$
Formally: $(\forall s > r) \lim_{n \rightarrow \infty} \frac{\llbracket P \rrbracket_d(n)}{n^s} = 0$

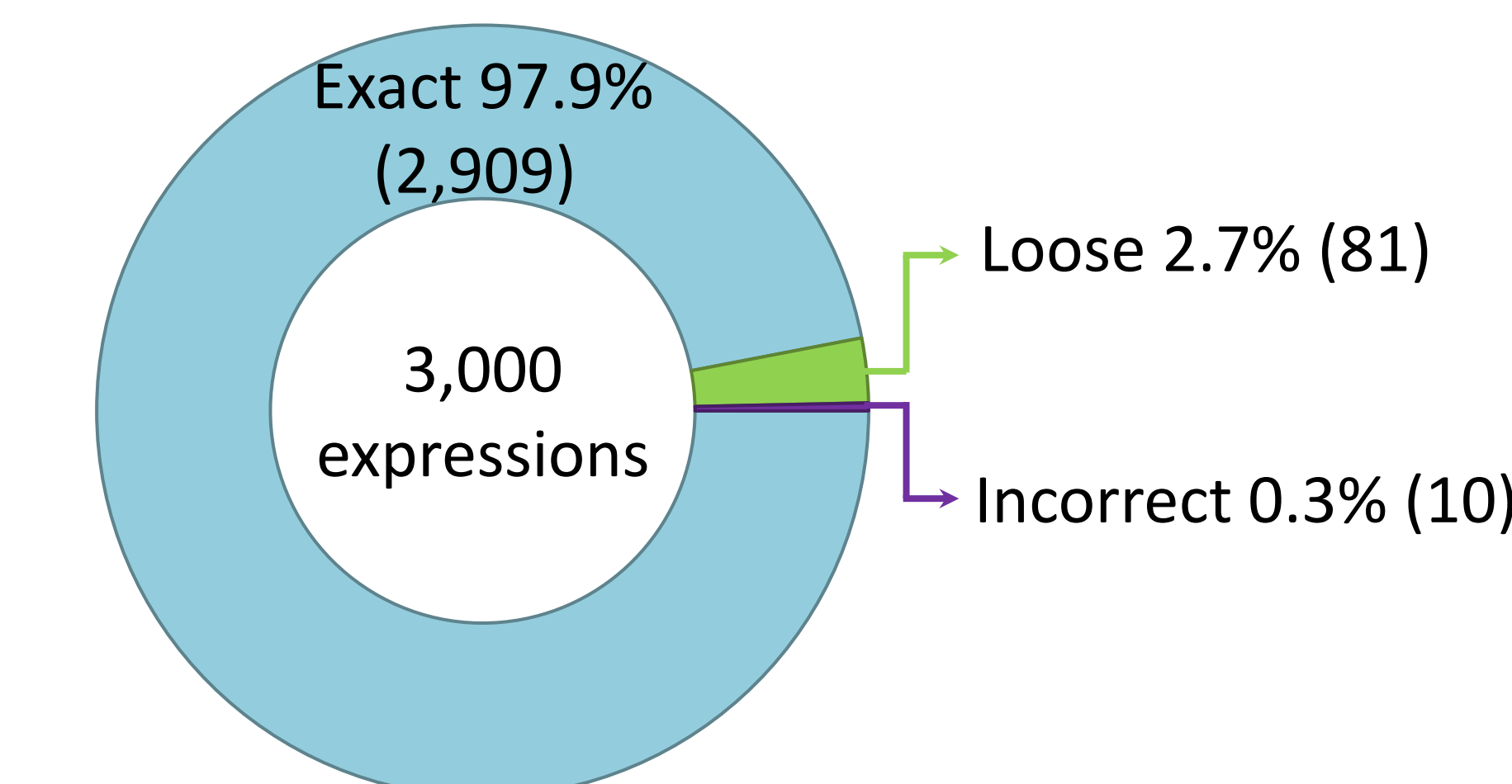
- ✓ Proof given in extended abstract
- ❖ Lean 4 mechanization in progress

Implementation

Lean 4 mechanization

- ✓ Semantic domains and orderings
- ✓ Ordering properties (reflexivity, etc.)
- ✓ Abstraction/concretization functions (noncomputable)
- ❖ Galois connection proof (ongoing)
- ❖ Assertion checking soundness proof (ongoing)

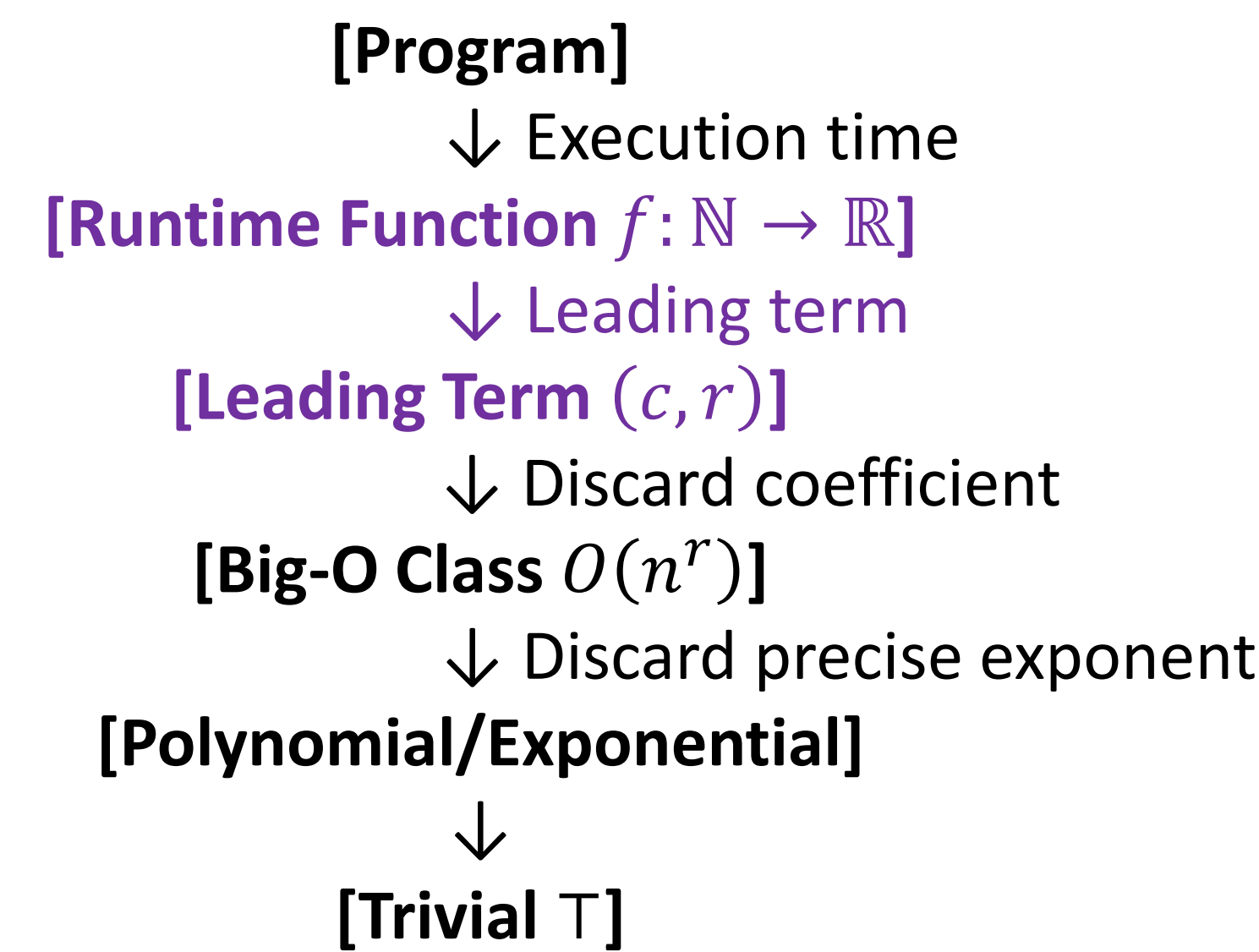
Interactive shell + fuzzer



- Exact (97.9%): LTA captures correct leading term
- Loose (2.7%)
 - Total cancellation: $n^2 - (n + n^2)$
 - Float precision: $n^2 + n^{2.000...1}$
- Incorrect (0.3%)
 - Float precision: $n^2 - n^{2.000...1}$

- ✓ **Empirical validation of soundness** within floating-point precision
- ✓ Interactive demonstration of **overapproximation**

Application: Runtime Complexity



- ✓ Factorizing runtime analysis
- ✓ Clear effects of each abstraction
- ✓ Tradeoff: precision vs. tractability
- ✓ Different purposes of each layer

Key Takeaways

- ✓ **Minimal**: No CPOs, no fixed points
 - ✓ **Complete**: Full Galois connection, sound assertion checking
 - ✓ **Accessible**: Familiar asymptotic analysis
 - ✓ **Pedagogical**: Low mathematical barrier to entry
- ⇒ **Essence of semantic approximation** without fixpoint machinery

Future work

- ❖ Complete mechanized proofs
(Galois connection & assertion soundness)
- ❖ Assessment of pedagogical effectiveness
- ❖ Extend to richer languages
 - Other constructs: $\log n, n!, \dots$
 - Loops, recursions, conditionals, ...
- ❖ Formalize floating-point semantics